Practical 21: Media APIs and Telephone APIs

Source: MODULE 2.pdf | Source pages: 84–88

Aim

To develop and test an Android application demonstrating media apis and telephone apis.

Steps to Perform the Practical

1. Create a project for the media and telephone API practical.

2. Add the controls and layout elements shown in the source practical.

3. Implement the supplied media API code for the selected media operation.

4. Implement the telephone API action using the Intent and URI shown in the source code.

5. Add and request any permissions that are explicitly required by the chosen operation.

6. Run the app and test each feature. Use a real device for telephone-related behavior when the emulator cannot provide it.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

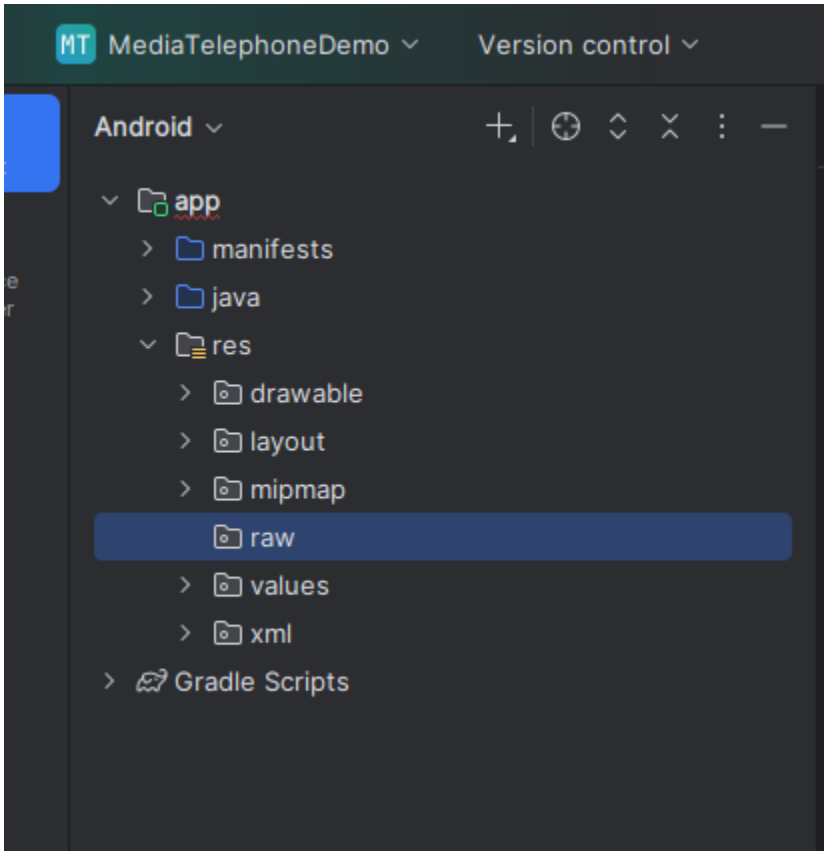
14. Develop an Android application implementing Media APIs and Telephone APIs.

Project Name: MediaTelephoneDemo

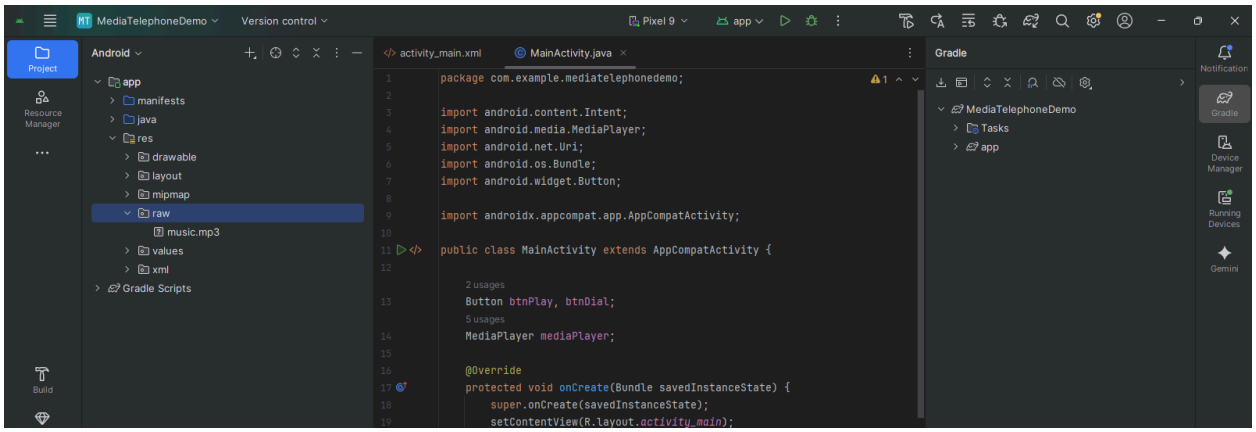
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
       android:id="@+id/btnPlay"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Play Music"/>
   <Button
       android:id="@+id/btnDial"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Open Dialer"
       android:layout_marginTop="20dp"/> </LinearLayout>
```

Add Audio File.
Create New folder in Android Project Panel: res raw



Download or use pre-existing small size MP3 file inside raw folder: music.mp3



MainActivity.java.

```
package com.example.mediatelephonedemo;
```

```
import android.content.Intent;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   Button btnPlay, btnDial;
   MediaPlayer mediaPlayer;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       btnPlay = findViewById(R.id.btnPlay);
       btnDial = findViewById(R.id.btnDial);

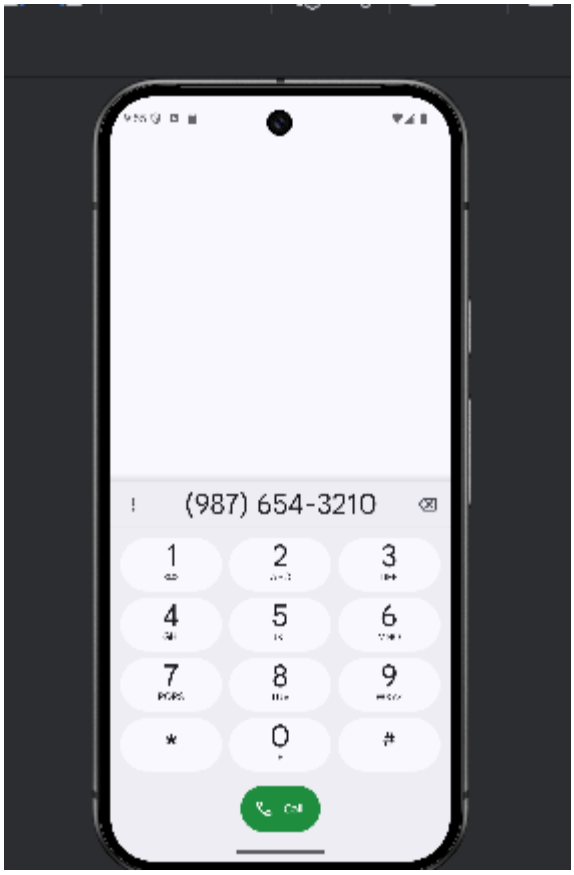
       // Media API
       mediaPlayer = MediaPlayer.create(this, R.raw.music);
```

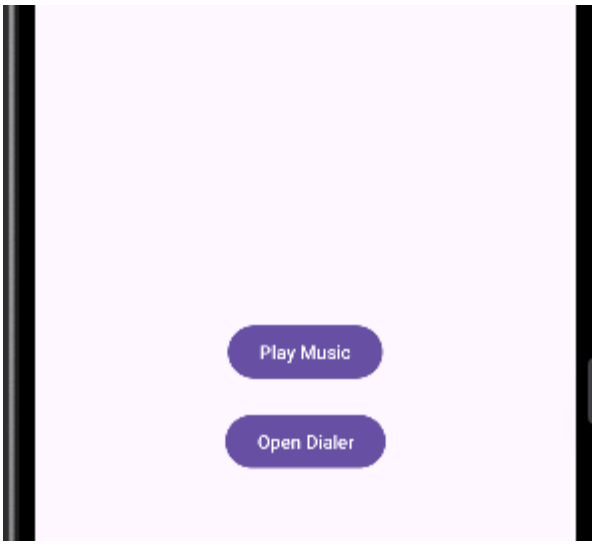
```
        // Play music
        btnPlay.setOnClickListener(v -> {
            mediaPlayer.start();
        });

        // Open phone dialer
        btnDial.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210"));
            startActivity(intent);
        });
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();

        if (mediaPlayer != null) {
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}
```


(987) 654-3210
1
2
3
4
5
6
7
8
9
*
0
#


Play Music
Open Dialer